Zach Siegel

Full-Stack Engineering | Data Science

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GitHub | Portfolio

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I'm a proficient and enthusiastic application builder passionate about optimization, automation, and statistics-driven decision-making. Always eager to employ mathematical programming.

Education

University of California, Los Angeles *MS - Operations Research. GPA 3.94* *2019-2021*

Pomona College *BA - Mathematics, Computer Science minor. GPA 3.63* *2010-2014*

Awarded **"Llewellyn Bixby Mathematics Prize"** 2014: to the student with highest achievement in the Mathematics dept.

Professional Experience

Full-Stack Data/AI Engineer *Current AI | Walleye Capital* *March 2024 - Present*

End-to-end product work, from ETL to UI. Deployed bulk ingestion pipelines for millions of records per day; synced data across transactional and analytical storage; designed customer-facing features.

Grounded generative AI. Built team's evaluation infrastructure to align AI-generated content with SME preferences, including a custom prompt optimization framework; a robust *named entity resolution* service; point-in-time Excel agent harness evaluation.

Innovated on tooling. Used a custom agentic loop to generate thousands of unique configurations to scrape high-quality structured data (without AI) from different websites each day.

Centered software rigor. Maintained high test coverage for backend and frontend Python and Typescript; extended testing paradigms to evaluate nondeterministic AI features. Conducted dozens of technical screens.

Won company-wide hackathon. Placed first of 48 teams with a resume-screening platform that aligns to manager preferences via annotations of synthetic data and automated prompt optimization.

Python•Typescript | MySQL•BigQuery•S3 | AI Harness Evals
Testing•CI/DevOps•Observability | Classification•NER | Full-Stack•Data Eng•AI Eng

Software Engineer & Data Scientist *Capsida Biotherapeutics, Inc.* *January 2022 - March 2024*

Led widely-used internal software platform (AWS+React+Django) providing: customized genomic design tools; long-running bioinformatics pipelines; scheduling automation and visualization; executive BI analyses; automated handoffs across platforms; integrations with lab robots; IP-related data mining. Onboarded cross-functional contributors, maintained testing suite, documented, and socialized tools.

Automated lab data plumbing by connecting platforms including: lab information database (Benchling), project management software (Smartsheet), and networked lab instruments. Built a QR-based inventory update system. Fully automated several lab instrument data ingest operations.

Developed forecasting models for primate immunogenicity, with a decision-support tool that anticipates spontaneous interruptions to animal availability (a widespread challenge in biotech). Built automated analysis pipeline that delivers increasingly precise recommendations to decision-makers as new assay results are recorded in lab information database. C suite relied on forecasts for (expensive!) contracts.

Python•Javascript | Redis•scikit-learn | AWS•On-Prem

Teaching Assistant for Graduate Students *Anderson School of Management, UCLA* *2020 - 2021*

Taught "Data and Analytics" to students in the MBA programs as part of PhD track teaching requirements.

Created (great) statistics course materials with a JupyterHub/git-based deployment pipeline. Students clicked a "magic link" to access cloud-provisioned, SSO-enabled compute environments.

Freelance Software Development

Automated Scheduling *Sinai Temple* *2018, 2019*

Automated and optimized scheduling Bar/Bar Mitzvah dates for ~130 students via mixed-integer linear programming.

Carpool Assignment Optimization *GroupThere - grouptherenow.com (offline)* *2017 - 2020*

Launched a carpool optimization tool for organizations. Minimizes total drive-time sum across groups of 2-100.

Community Safety Intervention Modeling *LA Community Action Network* *2017*

Re-implemented LAPD's "hotspot" generation algorithm. Compared hotspots to historical arrest/citation/crime report data from City of Los Angeles. Contributed results to community-generated report "Predictive Policing in Los Angeles".

Supply Chain Forecasting, Automation, and Optimization *FactoryOfEverything* *2016 - 2017*

Developed a model for purchasing, production, shipping, and holding over a factory-warehouse-retail system. Forecasting using classical signal processing, regression, and machine learning. Implemented MVP in MATLAB.

Python•Javascript•MATLAB | Flask•Angular | Pyomo•COIN-OR•Gurobi

Research

Pandemic Mitigation Optimization *Anderson School of Management, UCLA* *2021*

Optimizes decisions that affect compartment flow parameters in discrete-time SIRD disease progression model.

Fairness, Efficiency, and Feature-Awareness *Anderson School of Management, UCLA* *2020*

Extends strategies for algorithmic fairness from machine learning community to resource-allocation optimization setting.

Generative Models and Sparse Coding *Department of Mathematics, Pomona College* *2014*

Formalizes connections between *Boltzmann Machine* Distribution and unsupervised learning via sparse coding.

Anomaly Detection Using Dictionary Learning *University of Minnesota, Minneapolis* *2013*

Unsupervised anomaly detection in video data using dictionary learning and sparse coding. An NSF-funded REU. Awarded **"Outstanding Presentation Award"** at Joint Mathematics Meeting, 2014: top 15% undergraduate groups.